



SQRT Mission – Close Out Review

Flight Assessment of a High-Altitude Balloon Payload for Flash Evaporation Analysis Under Stratospheric Conditions.

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ABSTRACT

The SQRT mission was a high-altitude balloon experiment designed to investigate the thermal evolution of water undergoing flash evaporation under low-pressure stratospheric conditions. The experiment aimed to trigger rapid phase change at pressures of approximately 35mbar by releasing a water sample into a vented low-pressure chamber during ascent. The mission was motivated by both planetary science applications, including cryovolcanic plume formation on icy moons such as Enceladus and Europa, and terrestrial engineering applications involving reduced-pressure evaporation systems.

The payload was launched within a triple TupperSat payload aboard a stratospheric balloon from Armagh Observatory and Planetarium. The onboard systems included environmental sensing hardware, GPS telemetry, a triggered syringe-and-valve release mechanism, and an infrared thermal sensor intended to monitor transient thermal evolution during the planned flash evaporation event.

Although the payload experienced premature power loss before reaching the intended trigger altitude, the mission still successfully demonstrated payload integration, launch operations, telemetry acquisition, flight tracking, and recovery under real near-space conditions. Recovered telemetry data allowed reconstruction of the flight profile, including pressure evolution, thermal behaviour, ascent and descent dynamics, and trajectory tracking. Analysis of the flight data showed generally nominal high-altitude balloon behaviour, with measured ascent and descent profiles broadly consistent with pre-flight expectations.

While the primary scientific objective could not be fully achieved due to the early payload shutdown, the mission provided valuable engineering insight into systems engineering and integration, thermal management, and operational reliability within near space environments. Overall, the SQRT mission provided an invaluable opportunity to gain experience in mission lifetime operations, hands-on understanding of ESA-like Review Documentation and also lessons in the challenges and benefits associated with performing thermodynamic experiments using low-cost high-altitude balloon platforms.

1. INTRODUCTION

1.1 Mission Scientific Context and Justification

The SQRT mission was designed to investigate the thermal evolution of water undergoing flash evaporation when suddenly exposed to a low-pressure and low-temperature environment. This process is scientifically relevant because it demonstrates a fundamental thermodynamic response of liquids when the surrounding pressure drops below the saturation vapour pressure of the liquid itself. Under these conditions, the water rapidly boils, absorbs latent heat, and cools quickly, often producing plume-like vapour structures. This rapid phase transition and cooling behaviour was the core physical process the mission aimed to observe and analyse.

The scientific motivation for the mission comes from both planetary science and terrestrial engineering applications. In planetary science, similar rapid depressurisation processes are believed to contribute to the formation of cryovolcanic plumes on icy moons such as Enceladus and Europa, where volatile-rich liquids may become exposed to near-vacuum conditions through fractures in the surface. While the SQRT mission does not attempt to recreate the full geological complexity of these environments, it focuses on one of the key physical drivers behind these plume systems: the thermal behaviour of water during sudden exposure to reduced pressure. On Earth, the same physical principles are also relevant to multistage flash desalination systems, where water is intentionally exposed to lower pressures in order to drive rapid evaporation.



Figure 1. Cryovolcanic plume inspiration for SQRT.

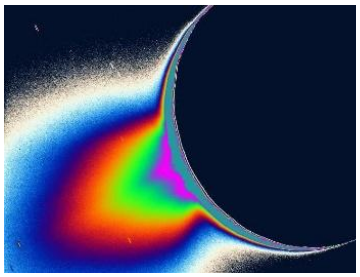


Figure 2. Flash evaporation thermal behaviour.



Figure 3. Industrial flash evaporation desalination systems.

A high-altitude balloon provided a particularly suitable platform for this investigation because the stratospheric environment naturally offers the low pressures required for flash evaporation to occur, while still allowing the experiment to remain lightweight, recoverable, and relatively low-cost. By using the TupperSat platform, the mission aimed to study this behaviour within a real time-varying atmospheric environment rather than relying entirely on expensive laboratory vacuum testing. The SQRT payload therefore used a triggered syringe-and-valve exposure system combined with onboard thermal sensing in an attempt to observe the transient cooling response of water during sudden depressurisation. Overall, the SQRT mission demonstrated that a relatively simple and compact balloon payload can still be used to investigate complex thermodynamic behaviour in realistic near-space conditions. Although

the payload experienced an early power loss and did not successfully reach the planned trigger event, the mission still demonstrated the successful integration, launch, flight, and recovery of a complex experimental TupperSat platform. The project provided valuable engineering insight into operating transient thermodynamic experiments in the stratosphere and highlighted important lessons for future low-cost high-altitude missions.

2. METHODS

2.1 Overview of the measurement and experimental principle

The SQRT mission was designed to investigate the thermal evolution of water exposed to low-pressure stratospheric conditions during high-altitude balloon flight. The experiment was based on the process of 'flash evaporation', where liquid water rapidly changes phase once the surrounding pressure falls below its saturation vapour pressure. This process absorbs latent heat, causing rapid cooling and potentially producing vapour or condensation structures. To investigate this effect in flight, the payload carried a specially printed removable cap-covered syringe-like water container which was connected to a vented low-pressure observation chamber through a servo-actuated pinch valve tubing system. The release mechanism was designed to trigger automatically at approximately 35mbar which is positioned at an altitude of approximately 23km, exposing the water sample directly to near-space atmospheric conditions. The primary scientific experimental instrument onboard the payload was the infrared thermal sensor, designed to be positioned within the observation chamber to monitor the thermal evolution of the water sample before, during and after the planned release event. Acquired housekeeping telemetry included altitude, pressure, temperature (both internal and external), GPS position, and timestamps, allowing the environmental conditions during flight to be correlated with the thermal measurements. The experiment therefore aimed to compare the thermal behaviour of the system before and after the trigger event in order to identify rapid cooling associated with flash evaporation.

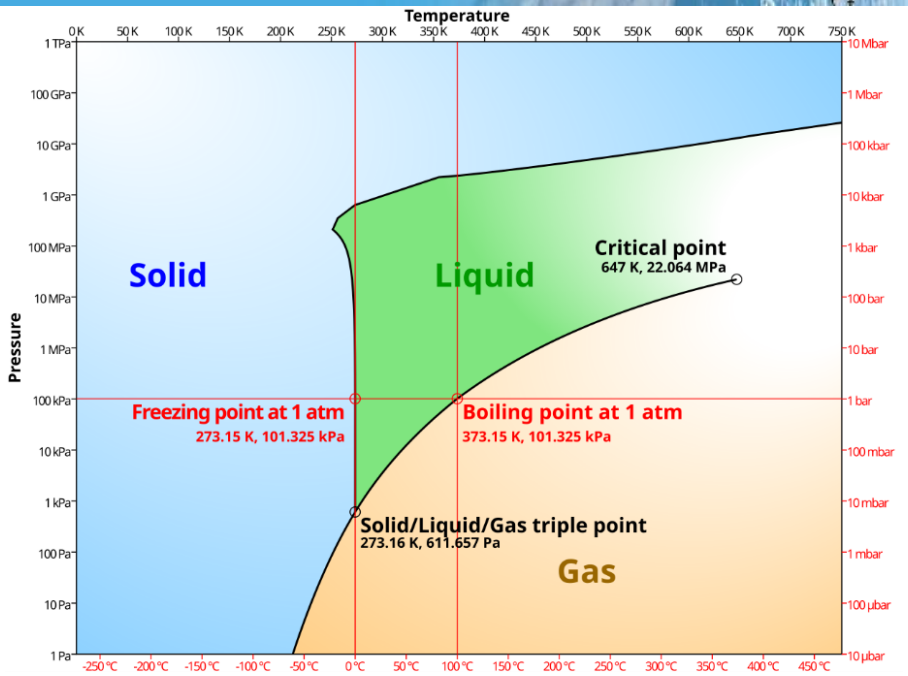


Figure 4. Phase Diagram of Water

Although the payload experienced premature power loss before the trigger conditions were reached, the mission still demonstrated the use of a high-altitude balloon platform for investigating transient thermodynamic processes under realistic stratospheric conditions. Thankfully, the payload was supported was additional telemetry provided by the SkyNet mission team which is later examined within the '***3.*** RESULTS' section.

2.2 Payload Design

The SQRT payload was designed to investigate the thermal evolution of water exposed to low-pressure stratospheric conditions while remaining within the mass and operational constraints and limits as per the *UCD-PRD-2026-0* document. The overall design focused on producing a lightweight, recoverable, and self-contained experiment capable of storing a water sample during ascent before releasing it into a vented low-pressure chamber at high altitude. The payload used a two-compartment internal layout to separate the experimental subsystem from the avionics and power systems, reducing the risk of water or condensation interfering with the electronics during flight. The main structure consisted of a reinforced Tupperware enclosure supported using threaded rods and external mounting hardware to improve structural stability throughout launch, ascent, burst, descent, and recovery. The experimental subsystem consisted of a capped syringe-based water reservoir connected through silicone tubing to a vented observation chamber. A servo-actuated pinch valve mechanism was used to isolate the water sample during ascent before releasing it once the trigger conditions had been reached. The observation chamber was vented to the external atmosphere to expose the water directly to near-space pressure conditions and attempt to initiate flash evaporation.

The avionics system was centred around a Raspberry Pi Pico microcontroller responsible for telemetry handling, environmental sensing, trigger logic, and onboard data logging. Supporting sensors measured altitude, pressure, temperature, and GPS position throughout flight. The primary scientific instrument was the MLX90640 infrared thermal array sensor, which was intentionally positioned with a bottom corner optimally wide angled view of the observation chamber to monitor the thermal evolution of the water sample following the trigger release.

Thermal management was an important part of the payload design due to the extremely low temperatures expected during stratospheric ascent. Internal insulation included fire blanket and insulating materials surrounding parts of the payload structure, while two handheld heating pads were also added to retain heat and liquid state during flight. Wiring and internal components were also mechanically secured to reduce movement and vibration-related disturbances throughout the mission. The internal payload within the electronics bus compartment consisted of a vaguely three tier design; on the TupperSat base the water chamber and servo motor sat securely positioned, above which the powerpack was supported by a custom printed holder, atop this was the electronics bay and accompanying electronic wiring and SD storage card, all of which were protectively contained within a printed electronics cap. See '*Figures 5* & *6*' below for visual CAD reference.

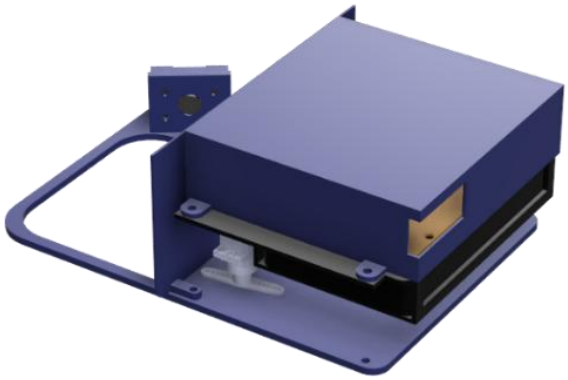


Figure 5. SQRT payload CAD model.

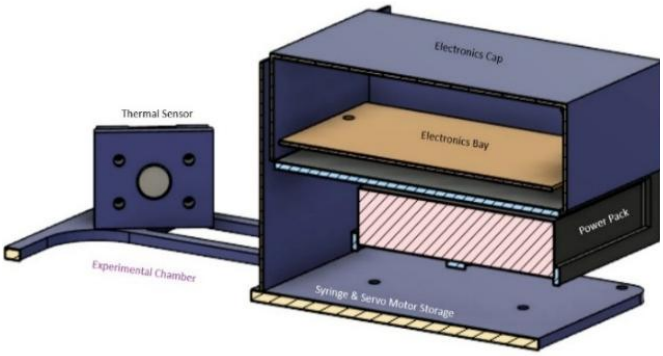


Figure 6. SQRT payload internal layout.

2.3 Description of payload as flown;

The flown SQRT payload consisted of a compact TupperSat platform contained within a reinforced Tupperware enclosure and supported using four threaded corner rods with external mounting rope for attachment to the balloon flight. Externally, further insulation was added in late phase design and was taped to the external plastic of the Tupperware box electronics and water container compartment. Internally, the payload was separated into two primary sections. The electronics section contained the Raspberry Pi Pico microcontroller, power pack, SD card storage, GPS module (which sat above the electronics cap and extruded through a hole on the Tupperware 'ceiling' cap), telemetry systems, and environmental sensing hardware. The second section formed the experimental observation chamber, which was vented to the external atmosphere and acted as the low-pressure environment for the flash evaporation event.

The flown experimental subsystem consisted of a syringe-based hot water reservoir connected through silicone tubing to the observation chamber. A servo-actuated pinch valve surrounded a small section of the tubing assembly in order to isolate the water sample during ascent and release it once the predefined trigger conditions had been reached. The water system was positioned within the payload to minimise leakage risk and reduce direct interaction between the liquid subsystem and the avionics. The primary scientific instrument onboard the payload was the infrared thermal sensor, which was mounted appropriately within the observation chamber to monitor the thermal evolution of the water sample following release. Onboard housekeeping systems measured altitude, pressure, external temperature, GPS position, and telemetry status throughout flight. All of the data taken were intended to be stored locally to an SD card while some telemetry data packs were intended to be transmitted to the ground station during ascent and descent.

Additional insulation materials and internal thermal barriers were incorporated into the flown payload to help reduce the effects of low external temperatures on both the electronics and water sample during flight. Internal components and wiring were mechanically secured prior to launch in order to reduce movement caused by vibration, ascent dynamics, burst, and landing impacts. Although the payload experienced an early power loss during flight and didn't successfully undergo the planned trigger event, the overall payload structure survived launch, ascent, descent, and recovery perfectly intact. Post-flight inspection showed no catastrophic structural damage to the external payload body following recovery.



Figure 7. Final payload configuration.

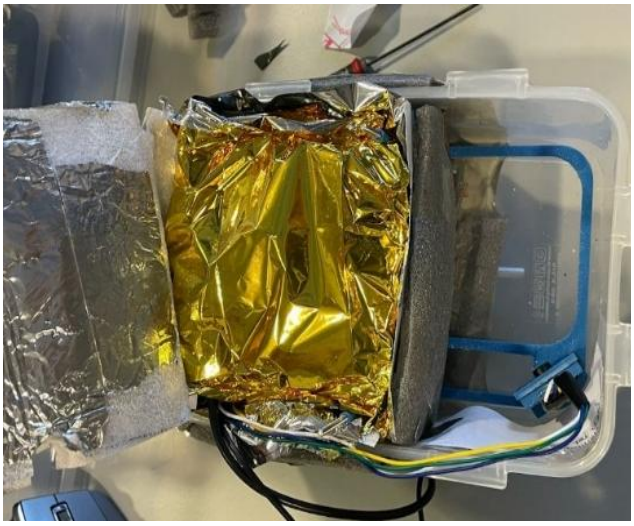


Figure 8. Internal payload assembly.



Figure 9. Flight-ready SQRT payload.

2.4 Flight and Launch Day Summary

The SQRT payload was launched aboard a high-altitude balloon as part of the TupperSat programme on the 23rd of April 2026 from the Armagh Observatory and Planetarium (AOP) launch site. The launch team departed UCD at approximately 07:30 and arrived at the launch site shortly before 11:00 following a fuel and rest stop at the M1 Service Station Northbound. After arrival, the launch area was prepared and a protective ground sheet was secured using weights and toolboxes before balloon inflation began. Protective gloves were worn by all personnel handling the balloon to reduce risk of accidental puncture during filling operations. In addition, high visibility clothing was worn by most onsite.



Figure 10. Balloon inflation at launch site.



Figure 11. Launch preparation operations.



Figure 12. SQRT mission team.

Balloon inflation began at approximately 11:22 and environmental conditions at the launch site resulted in a noticeable wind interaction with the balloon during filling. At approximately 11:27 the balloon began lifting unpredictably due to wind loading final inflation was completed. During filling operations, at approximately 11:30 the balloon reached an internal pressure of approximately 50 mbar and a second helium cylinder was connected shortly afterwards to complete inflation.

The SQRT payload powerpack was powered on at approximately 11:31 and underwent final assembly and pre-flight verification checks prior to launch. By 11:40, the ground station systems had successfully connected and were receiving telemetry from the payload. The TupperSat payloads were then attached to the balloon harness assembly shortly before 12:00 as spectators and members of the public gathered at the launch site to observe the launch event.



Figure 13. Ground station setup.



Figure 14. Final payload checks.



Figure 15. Payload harness assembly.

The balloon launch occurred at approximately 12:06, with visual line-of-sight to the payload being lost by approximately 12:08 shortly after ascent. Recovery operations began almost immediately, with the second recovery vehicle departing at approximately 12:13 to track the predicted landing location using the live telemetry data. At approximately 12:33, 27 minutes after launch, telemetry was successfully received from directly beneath the flight path near Moy, close to Charlemont. Multiple elevated search locations were then used throughout the chase to visually reacquire the balloon and payload during descent.

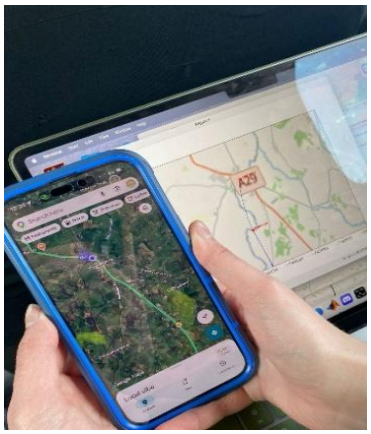


Figure 16. Recovery tracking operations [Timestamp: 00:27:00].



Figure 17. Visual payload tracking.



Figure 18. Final descent observation.

Telemetry and GPS data later suggested that the balloon had burst during the early afternoon descent phase. At approximately 13:28, the descending balloon and payload were visually identified using binoculars from the final search viewpoint location. The payload touchdown was observed from the road shortly afterwards and the SQRT payload was successfully recovered at approximately 14:35 at a location of 54°34'40.8"N, 6°46'28.9"W.

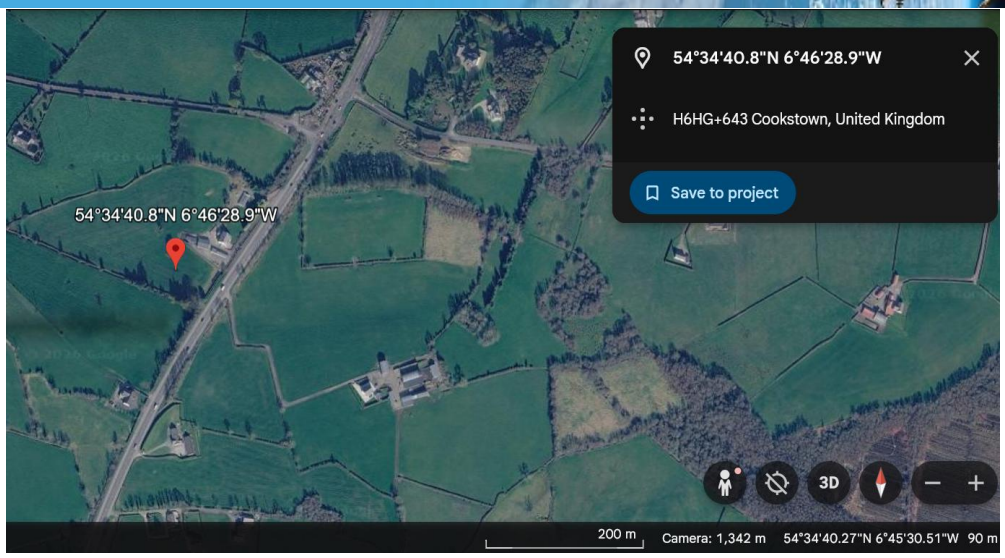


Figure 19. Exact payload recovery location.

Following recovery, the team travelled to Dungannon where the payload underwent initial inspection and telemetry review. Early inspection showed that the water sample remained inside the experimental chamber, indicating that the planned trigger event had not successfully occurred during flight. Preliminary analysis of the telemetry suggested that the payload experienced an unexpected power failure shortly after launch. Initial assumptions focused on possible hardware issues such as loosened wiring or connector failure during ascent, although later post-flight analysis suggested that thermal conditions and heavy insulation may also have contributed to the system failure. Further investigation of the telemetry and payload performance is discussed in later sections of this report. Despite the incomplete execution of the primary scientific objective, the launch, chase, tracking, and recovery operations were carried out successfully, providing valuable launch and land day operational experience and post-flight engineering insight for the SQRT mission team.

2.5 Additional Launch Day Information;

2.5.1 Launch Day Pre-Launch Preparation Steps;

Preparation for the SQRT launch began prior to arrival at the launch site through review of the predicted balloon flight path and estimated landing region using the provided trajectory prediction tools outlined below. Particular attention was given to expected wind direction, projected landing location, road accessibility, and possible recovery difficulties within the predicted landing region. Recovery routes and major road connections were reviewed beforehand to improve the efficiency of the chase and retrieval phase following launch.

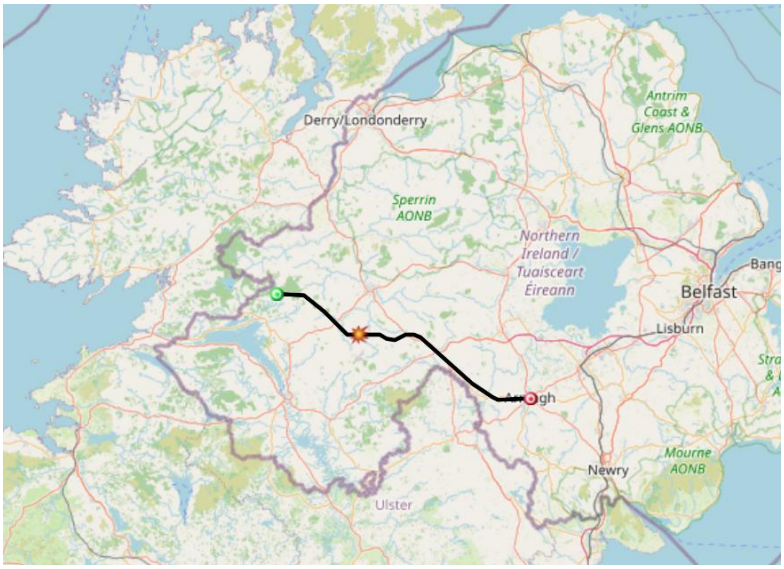


Figure 20. Early predicted flight trajectory using https://predict.sondehub.org/

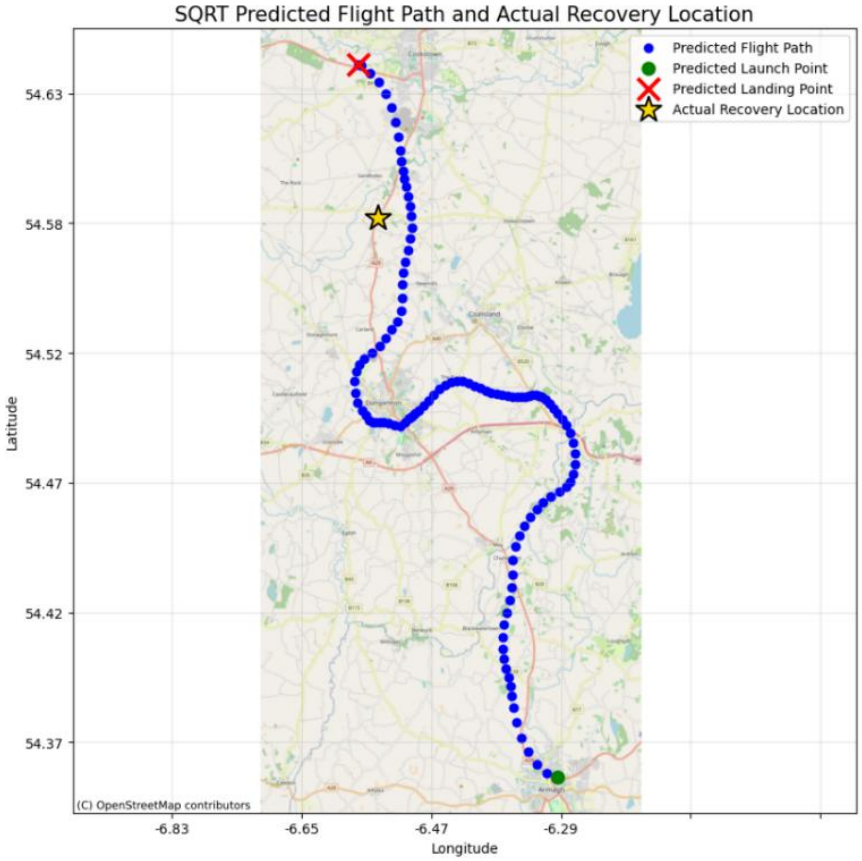


Figure 21. Predicted flight path updated early morning of launch day.

As part of the recovery preparation, I as a designated driver, also reviewed the expected drive path from the UCD to the launch site, and from there towards the projected landing area, including possible stopping locations, elevated viewpoints, and accessible roads suitable for visual tracking and recovery operations. Communication between chase vehicles and the ground station team was also planned prior to launch to ensure continuous coordination during ascent and descent tracking.

Prior to launch, the SQRT payload underwent final pre-flight checks including powering on the system, confirming telemetry reception through the ground station, verifying sensor operation, inspecting the valve and tubing system, and ensuring the payload remained securely assembled within the flight structure. Once the balloon inflation was complete, the payloads were attached to the balloon harness assembly and a final visual inspection was carried out before launch authorisation was given for connection to the ballon harness.

2.5.2 Some useful launch site notes…

The Armagh Observatory launch site was within its 'Stone Calendar Circle' which sits at a height of approximately 60m above sea level with coordinates of 54°35'35.7"N, 6°65'07.2"W.

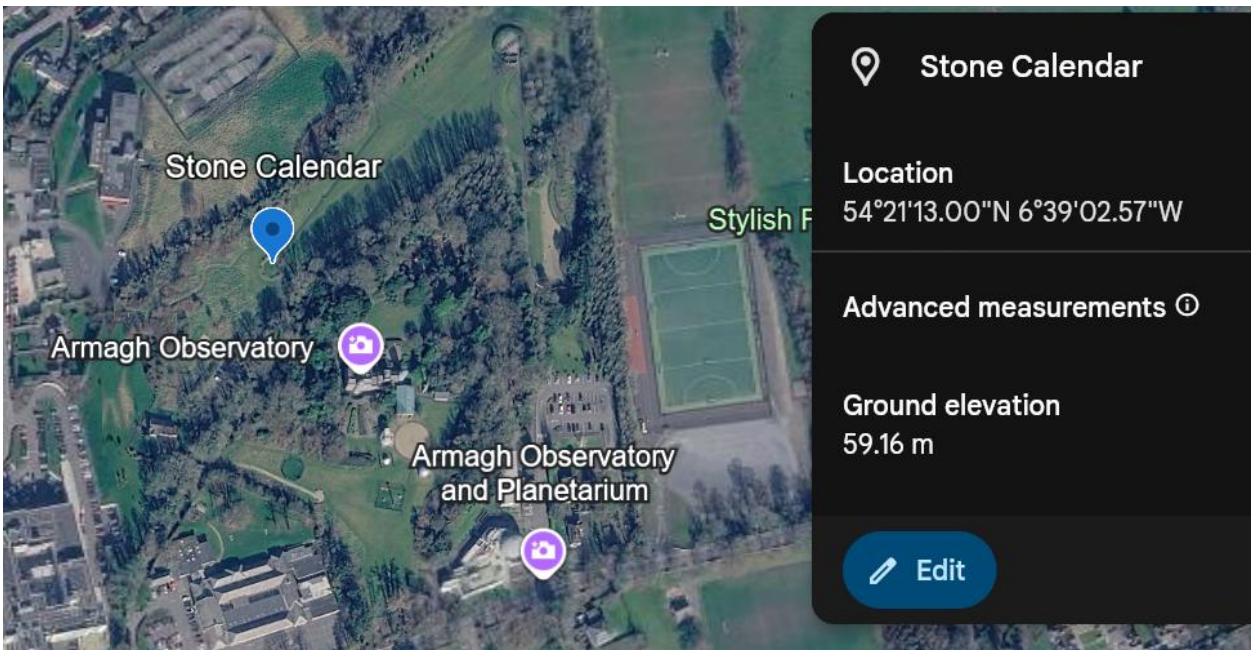


Figure 22. Launch Site Location

Below details the site weather conditions as recorded by Armagh Observatory: https://climate.armagh.ac.uk/contents.html

- Observer comments; *Sunny before observation. Blue sky and detached clouds at observation 9:30am, cool breeze, sunny interval to sunny day*
- *Weather round-up for April published by Mark Bailey of AOP; "The sunniest day was the 23rd with 12.4 hours of strong sunshine"*

- Recorded Data by Automatic Weather Station for launch day; 23/04/2026
    - Pressure; 1030.5mb.
    - Dry Bulb temperature; 11.1°C.
    - Wet Bulb temperature; 8.7°C.
    - Wind Direction 90° East from North.
    - Wind Speed; 13 knots.
    - Total Synoptic Cloud Coverage; 5oktas.
    - Visibility from weather station; 6kms.
    - Present Weather; 2 - State of sky generally unchanged.
    - Rainfall; 0.00mm.
    - Sunshine; 12.4hrs.
    - Soil temperature at depth of 30cm; 11.1°C
    - Soil temperature at depth of 100cm; 10.4°C
    - Maximum Temperature; 17.2°C
    - Minimum Temperature; 6.8°C

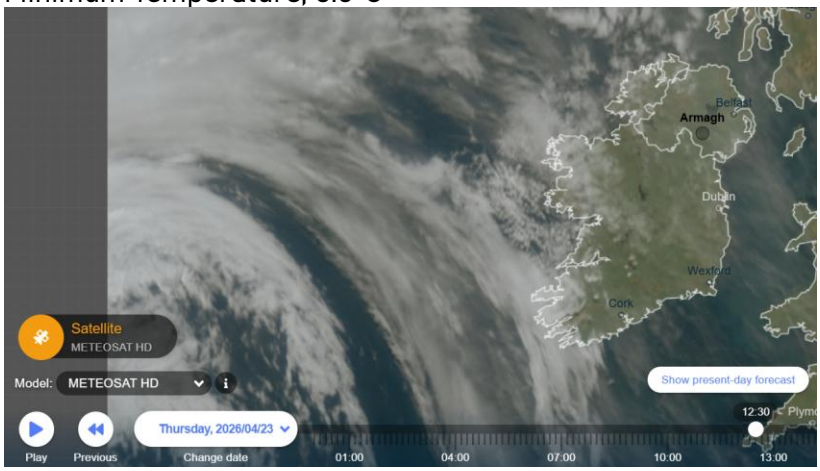


Figure 23. Launch day cloud conditions.

2.6 Launch Day Challenges

Although the overall launch day operations were successful, several practical and technical challenges were encountered throughout the launch day. One of the first issues occurred during the early launch site preparation, where the ground conditions proved somewhat unsuitable for securing the balloon ground sheet so we opted to attempt using pegs packed for grass based launch sites, unfortunately the pegs didn't appropriately fit the ground sheet and as a result, had to be discarded from use. Instead, toolboxes and other heavy equipment were used to weight down and secure the protective sheet during balloon inflation.

Weather conditions throughout the day were generally very good, with great visibility and dry conditions during launch and recovery operations. However, with the overall clear skies, we also experienced problematic and inconvenient wind conditions at the launch site which occasionally created difficulties while handling the balloon during inflation, as the

balloon almost lifted prematurely before final helium filling and integration of the TupperSat harness had been fully completed. While these conditions ultimately remained manageable, stronger winds could have posed a far more serious operational risk during launch preparation.

During flight operations, intermittent GPS and telemetry reception also became a challenge. Telemetry reception from both the SQRT and RadSat payloads were lost for significant portions of the flight, although the SkyNet payload transmissions continued to provide useful positional information during the chase and recovery phase. Maintaining line-of-sight tracking before landing became increasingly important towards the end of the descent phase, which placed a lot of additional time pressure on the recovery as our vehicle tried three different viewpoints in an attempt to position our line-of-sight optimally before touchdown.

Recovery itself introduced further practical difficulties due to muddy field conditions within the landing area, with movement across the terrain becoming difficult at times during retrieval operations. The field where the TupperSats landed, although easily accessible from the road, contained many mud-filled puddle points and it was difficult to walk through without getting stuck. Despite this, the payload was successfully recovered without much structural damage. It was noticed later that the two Go-Pro cameras which accompanied the payload via a printed attachment may have become brittle and cracked or loose during the flight as they were not attached to the payload upon recovery. Further video imagery analysis post-flight supports this theory. This was an unfortunate outcome as the cameras would've provided invaluable data to support post analysis within this report. Nonetheless, this could be a lesson learnt for future missions.



Figure 24. Suspected camera pre-detachment moment.



Figure 25. Possible camera detachment moment.

Following recovery, initial inspection of the SQRT payload led to concerns that the water sample may have leaked during flight. However, more detailed post-flight inspection later confirmed that the water remained contained within the system and that leakage had not occurred. This concern was later assumed to be potential water vapour within the payload dur to the pressure and temperature differentials during flight.

In fact, avoiding premature freezing of the water sample had originally been considered one of the largest environmental risks to the mission due to the extremely low external temperatures expected during ascent. While the payload insulation appears to have successfully prevented freezing, later analysis and testing suggested that the insulation and thermal conditions may instead have contributed to elevated battery temperatures beyond their intended operational range of 0°C to 35°C.

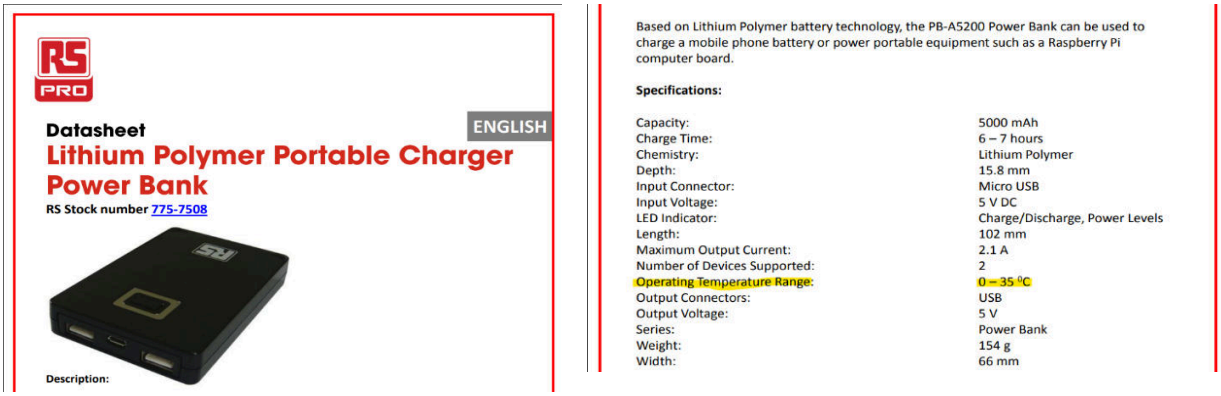


Figure 26. Battery operating limits. Note: operational temperature range in yellow.

The biggest challenge of the mission was, of course, the unexpected power failure experienced during ascent before the payload reached the required altitude and pressure conditions needed to trigger the experiment. As a result, the servo motor actuator pressure release trigger condition was never electronically met and therefore, wasn't initiated, as such the primary scientific objective of measuring the water's thermal response to near space conditions could not be completed during flight. This unfortunately was the biggest disappointment of the mission for the team, as it significantly reduced the usable experimental data available for post-mission analysis. This outcome was also hugely disheartening as we had worked extremely hard and long hours on the mission only for such a letdown of the trigger event never actually occurring, despite so many thermal tests!

Despite this, the mission still provided valuable operational, integration, launch, and recovery experience, along with important engineering lessons regarding thermal management and power system reliability in stratospheric environments, in addition to being an extremely fun experience. Although the flash evaporation event couldn't occur due to power loss, payload recovery did allow access to the onboard SD card data which had approximately 50 minutes of both usable housekeeping telemetry and usable visual thermal sensor data of the payload temperature shifts during flight. See '***3.*** RESULTS' for further analysis.

2.7 High Points / Successes

Despite the incomplete execution of the primary scientific objective, the SQRT mission still achieved several important operational and engineering successes throughout the project and full launch experience. Launch operations overall went fairly smoothly, with good SQRT

payload integration, balloon inflation, telemetry verification, and release carried out within the planned launch timeline. Communication between teams during launch and recovery was effective using a group Whatsapp including contact personnel for each vehicle, and the coordination between drivers, navigators, and trackers helped make the recovery an efficient chase and retrieval operation throughout the flight.

Early ascent telemetry remained stable following launch, allowing the team to monitor altitude, pressure, GPS position, and other essential housekeeping data during the initial flight phase. Although the payload later experienced a power failure, the telemetry systems functioned successfully during the early stages of ascent and provided some valuable flight information for post-flight analysis.

One of the major environmental risks that concerned us before launch was the premature freezing or leakage of the water sample due to the low temperatures expected in the stratosphere. Post-flight inspection confirmed that the water remained successfully contained and had not frozen within the system prior to the failure event, demonstrating that design mitigation through insulation and containment approach were at least partially successful under real flight conditions, even if a little too successful for their intended purpose.

The payload structure itself also survived launch, ascent, descent, landing, and recovery without any mechanical damage. This demonstrated that the overall structural design, mounting approach, and subsystem integration were capable of surviving the dynamic conditions experienced during a high-altitude balloon mission. Recovery operations were completed both successfully and very efficiently as the payload was visually tracked during descent before touchdown. The TupperSat was then recovered quickly after landing, allowing post-flight inspection and telemetry review to begin shortly afterwards whilst celebrating a successful launch and recovery.

To conclude, it's clear that the mission also provided invaluable experience in systems integration, launch operations, recovery coordination, telemetry handling, and real-time problem solving within a realistic near-space mission environment. Overall teamwork throughout the project and launch day remained very strong, particularly during the recovery and troubleshooting phases following the unexpected system failure. The SQRT team, class group and instructors as a whole, performed very well as a unit and proved that the teamwork involved was key to such a successful launch day.



Figure 27. Full mission team post payload recovery.



3. RESULTS

3.1 Flight Telemetry Overview

This section presents the housekeeping telemetry collected during the SQRT mission flight operations. The available telemetry includes altitude, atmospheric pressure, GPS position, internal and external temperature measurements, trigger system status, and general subsystem health data recorded during the early stages of ascent. These measurements were used to evaluate the environmental conditions experienced by the payload, assess subsystem performance, analyse the flight profile, and compare the observed mission behaviour against the original pre-flight predictions.

Due to the unexpected onboard power failure experienced during ascent, telemetry transmission and onboard logging from the SQRT payload were interrupted before the planned trigger event altitude was reached. As a result, the available SQRT telemetry dataset primarily reflects the early ascent phase of the mission rather than the complete flight profile through burst and descent. Despite this limitation, the recovered telemetry and onboard SD card logged files still provided valuable insight into the operational behaviour of the full payload and the environmental conditions encountered during flight. Although SQRT telemetry was mostly lost, recovery operations continued to be supported using telemetry transmitted from the Skynet payload, which remained operational during later stages of the flight and assisted with trajectory tracking and payload recovery. This telemetry was kindly donated by the SkyNet team for mission analysis and as such is acknowledged with respect and gratitude.

3.2 Pressure Profile with Altitude

'Figures 30 and 31' below show the measured atmospheric pressure profile as a function of altitude during the early ascent phase of the SQRT mission. As expected, the recorded pressure decreased exponentially with increasing altitude, consistent with standard atmospheric behaviour in the lower atmosphere. This trend confirmed that onboard pressure sensing system operated well during the portion of flight for which telemetry remained available.

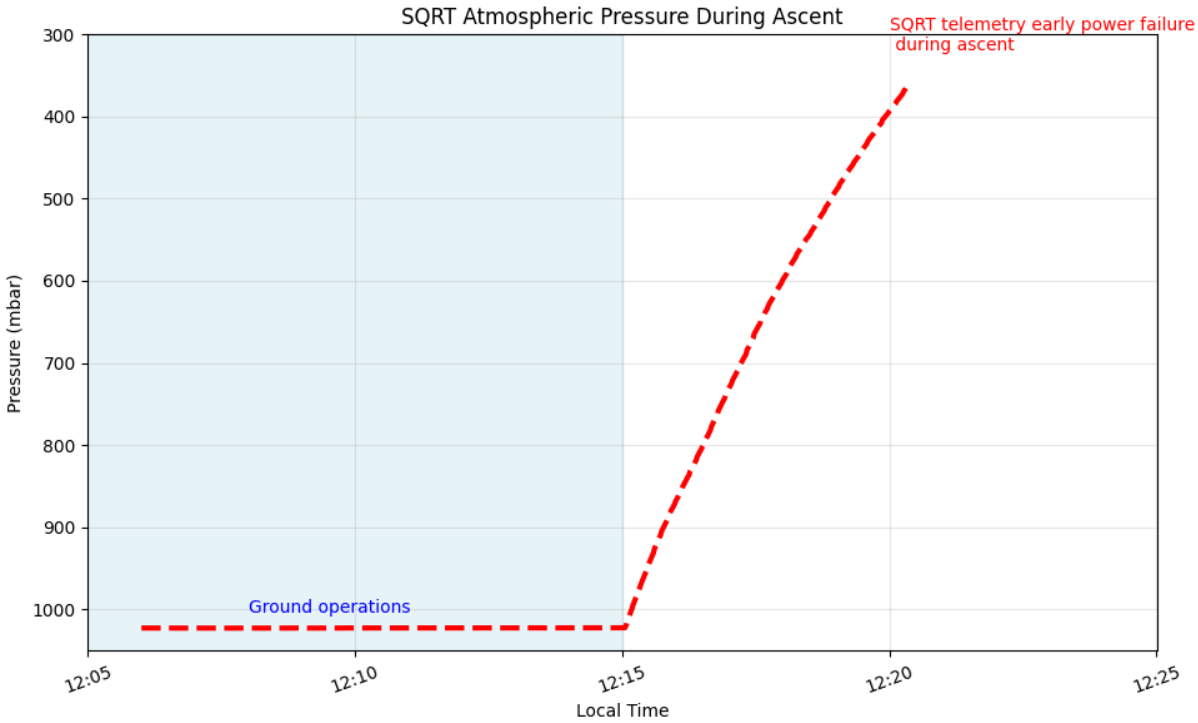


Figure 28. SQRT ascent pressure profile including ground operations segment prior to launch.

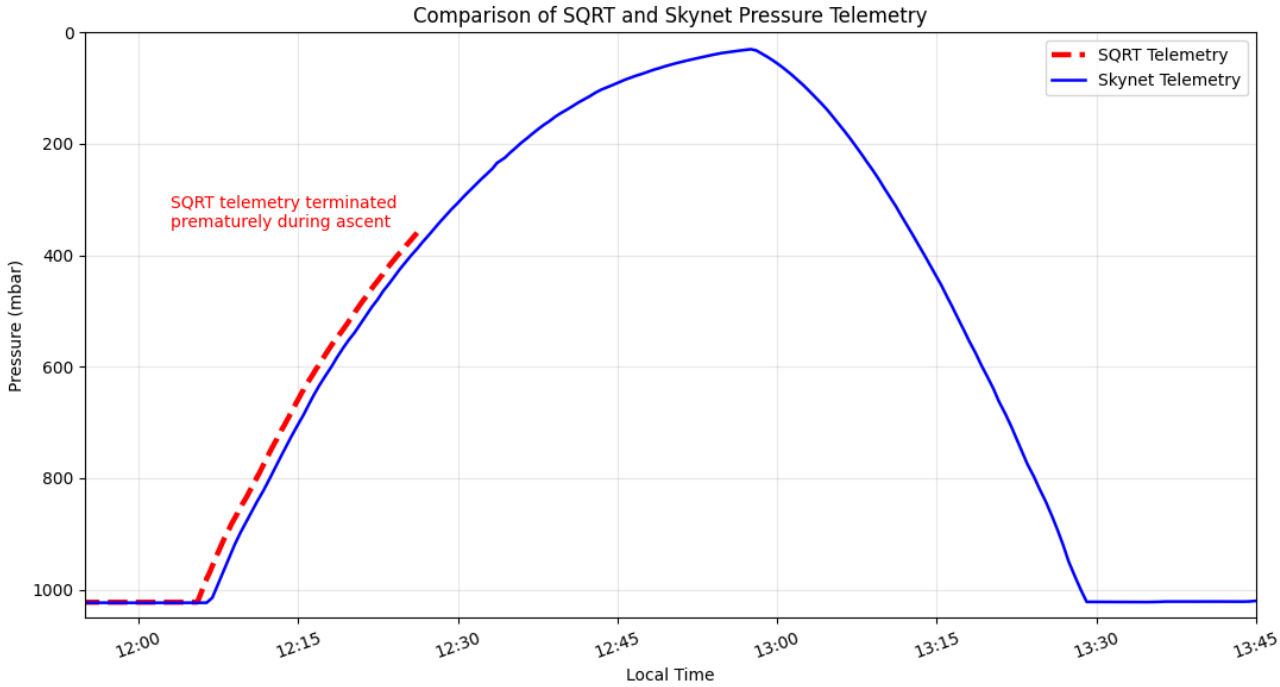


Figure 29. SQRT and Skynet pressure time series comparison.

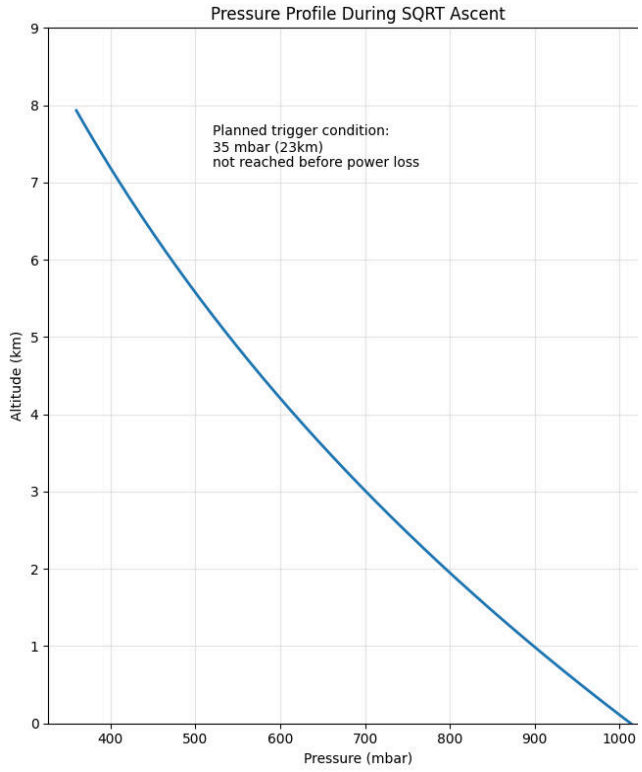


Figure 30. Pressure variation during ascent.

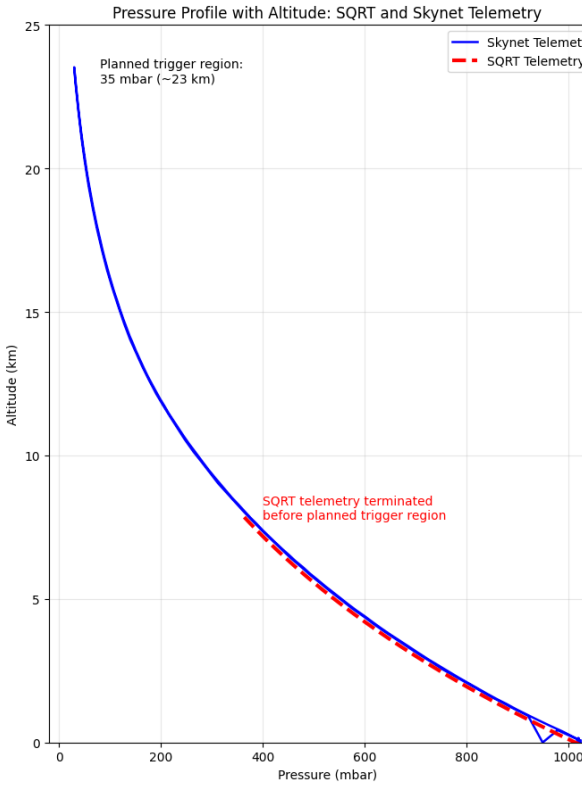


Figure 31. Pressure profile with altitude.

Monitoring the pressure profile was important for the mission because the planned flash evaporation experiment relied on the payload reaching sufficiently low external pressures before triggering the water release event. The trigger logic was designed around an approximate pressure threshold of 35mbar, corresponding to an altitude of roughly 23 km under standard atmospheric conditions. Comparison between the measured pressure trend and standard atmospheric expectations showed generally consistent behaviour during ascent, indicating that the environmental sensing subsystem remained functional prior to the onboard power failure. No major anomalies or physically unrealistic pressure measurements were observed within the available telemetry dataset. However, due to the premature loss of payload power, the complete pressure profile through the higher stratosphere, balloon burst, and descent phases could not be obtained from the SQRT telemetry alone. Although the planned trigger altitude region was never successfully electronically reached before the payload shutdown occurred, the available pressure telemetry still provided useful validation of the sensor subsystem and confirmed that the payload was ascending through progressively lower-pressure atmospheric conditions prior to failure.

3.3 Internal and External Temperature Profile

This section presents the measured internal and external temperature profiles recorded during the SQRT ascent. These temperature measurements were analysed as a function of altitude to examine the thermal behaviour of the payload under changing



atmospheric conditions. Special attention was given to the effectiveness of the payload insulation, the response of the external temperature sensor to stratospheric cooling, and the influence of internal electronic heat generation on subsystem temperatures. The external temperature profile generally followed the expected atmospheric trend during ascent, with temperatures decreasing as altitude increased through the troposphere. This behaviour was consistent with the reduction in atmospheric density and ambient thermal energy at higher altitudes. Although SQRT telemetry was cut-off before reaching the upper stratosphere, the recorded data still demonstrated the expected cooling behaviour during the early ascent phase and provided confidence in the operation of the external sensing subsystem.

In contrast, the internal temperature measurements exhibited significantly smaller variations throughout ascent. This indicated that the insulated payload structure along with the addition of the internal heating pads, successfully reduced the rate of thermal change experienced by the internal electronics and experimental hardware. Heat generated by the electronics bay and battery system most likely contributed to maintaining the internal compartment temperature above the external ambient conditions. This demonstrated that the payload retained high temperatures and internal thermal stability despite the rapidly cooling external environment. The temperature difference between the internal and external sensors provided an indication of the effectiveness of the payload insulation and thermal management strategy. Maintaining elevated internal temperatures was particularly important for battery reliability and electronic subsystem operation, as low temperatures can significantly reduce lithium battery performance and potentially contribute to voltage instability or power loss.

Although the mission experienced premature telemetry termination before reaching the intended trigger altitude, the available thermal data still provided useful insight into the thermal response of the payload during ascent. The measurements confirmed that the sensing systems operated successfully during the early flight phase and demonstrated the value of near-space balloon platforms for thermal and environmental experimentation under real atmospheric conditions.

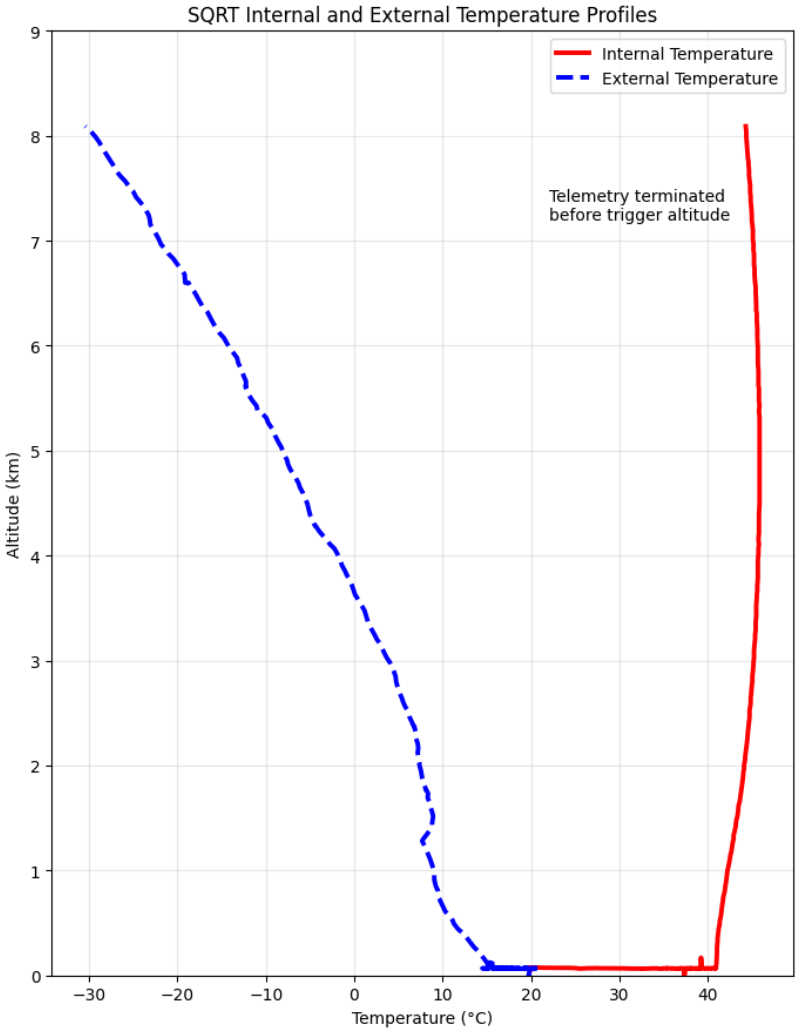


Figure 32. Internal and external temperature profiles.

3.4 Flight Groundtrack and Trajectory Prediction

The reconstructed groundtrack in '*Figure 33*' below shows a mainly south-westerly drift during ascent, which was consistent with wind conditions on launch day. The actual trajectory generally followed the predicted pre-flight model, although some deviations in the final descent region were observed due to local wind variations and uncertainties in burst altitude prediction. The launch occurred from Armagh Observatory, with the balloon initially drifting westward before transitioning southward during higher-altitude ascent. Following burst, the payload descended near enough to the predicted recovery region near Dungannon, which allowed us to successfully recover the payload and telemetry systems. Comparison with the pre-flight prediction demonstrated that the forecasting model provided a reasonable approximation of the overall flight path, although differences in wind speeds at different altitudes could have contributed to the trajectory differences.

The availability of continuous Skynet telemetry significantly improved situational awareness during the recovery phase, particularly after SQRT telemetry ceased mid-flight. Real-time GPS tracking enabled the recovery team to reposition effectively and visually observe the payload during final descent and landing.

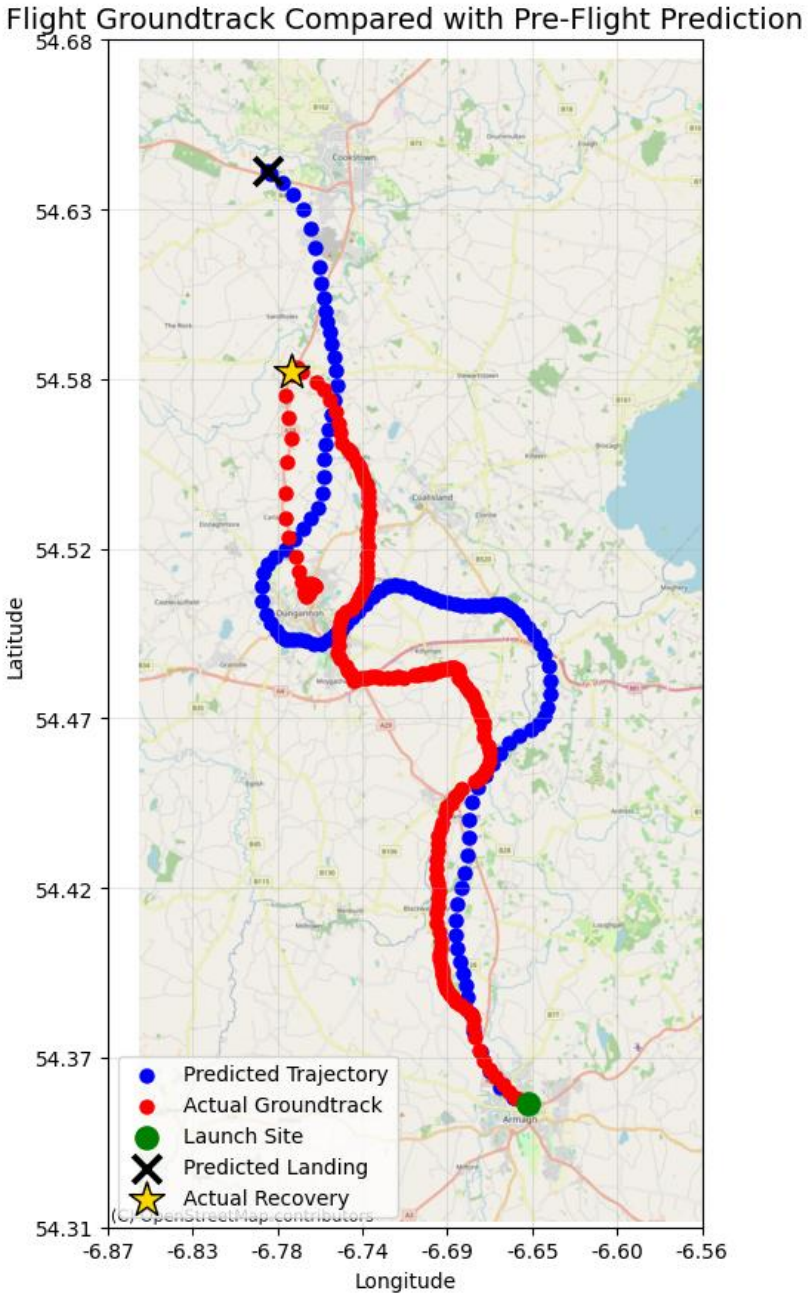


Figure 33. Flight groundtrack comparison.

3.5 Ascent and Descent Profile Analysis

The altitude–time profile provided an overview of the complete mission trajectory from launch through recovery. Using the SkyNet telemetry data, the flight profile clearly showed the major flight phases, including steady ascent, balloon burst, rapid descent, parachute stabilisation, and landing. During ascent, the payload demonstrated a near-linear increase in altitude with time, which shows a stable ascent rate throughout most of the climb phase. The average ascent rate was estimated to be approximately 6m/s, which was consistent with expectations for the balloon and payload configuration used during the mission. Following the burst, the telemetry showed a rapid descent phase before the parachute stabilised the payload at a lower terminal descent velocity suitable for recovery.

The full SkyNet telemetry dataset was particularly valuable for reconstructing the complete flight profile, especially following the premature termination of SQRT telemetry during ascent. Overall, the recovered telemetry demonstrated that the flight system behaved broadly as expected and provided useful insight into the ascent and descent characteristics of the mission

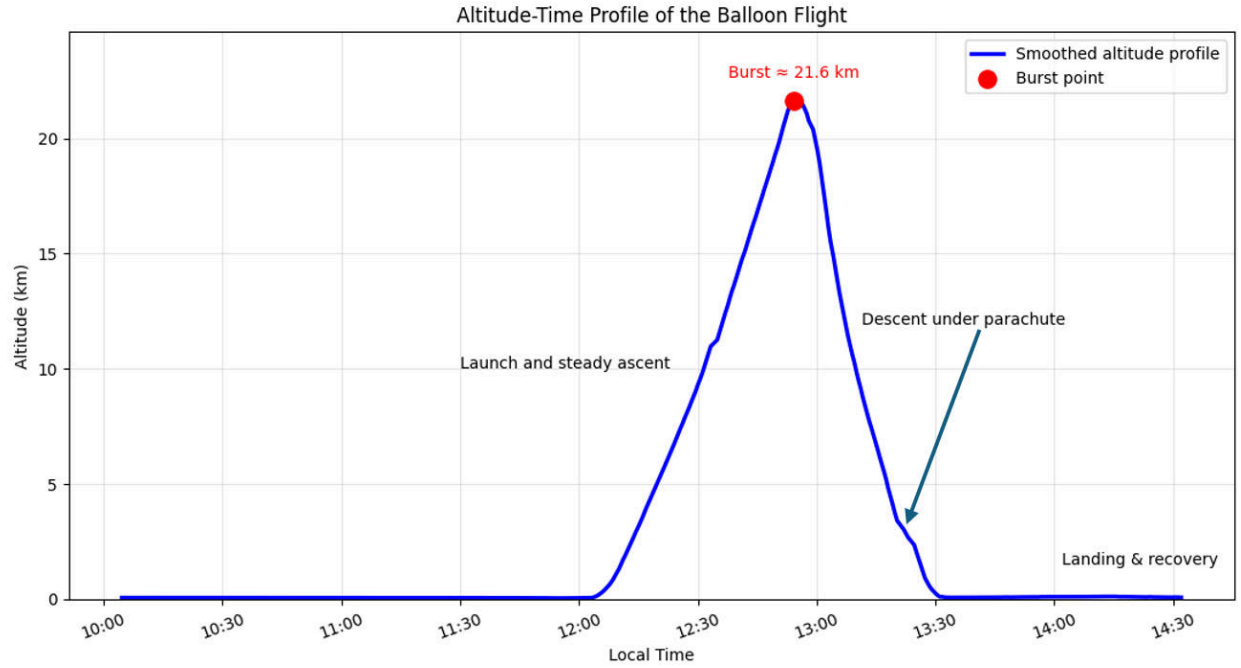


Figure 34. Altitude–time flight profile including balloon burst altitude noted in red.

Please see below table for all relevant calculated burst, descent and ascent parameters;

Table 1. Balloon burst, ascent and descent output parameters

Burst altitude: 21.62 km	Max descent rate: -15.09 m/s	Max ascent rate: 9.30 m/s
Burst time: 12:54:18	Average descent rate: -3.41 m/s	Average ascent rate: 4.05 m/s

3.5.1 Vertical Velocity Analysis

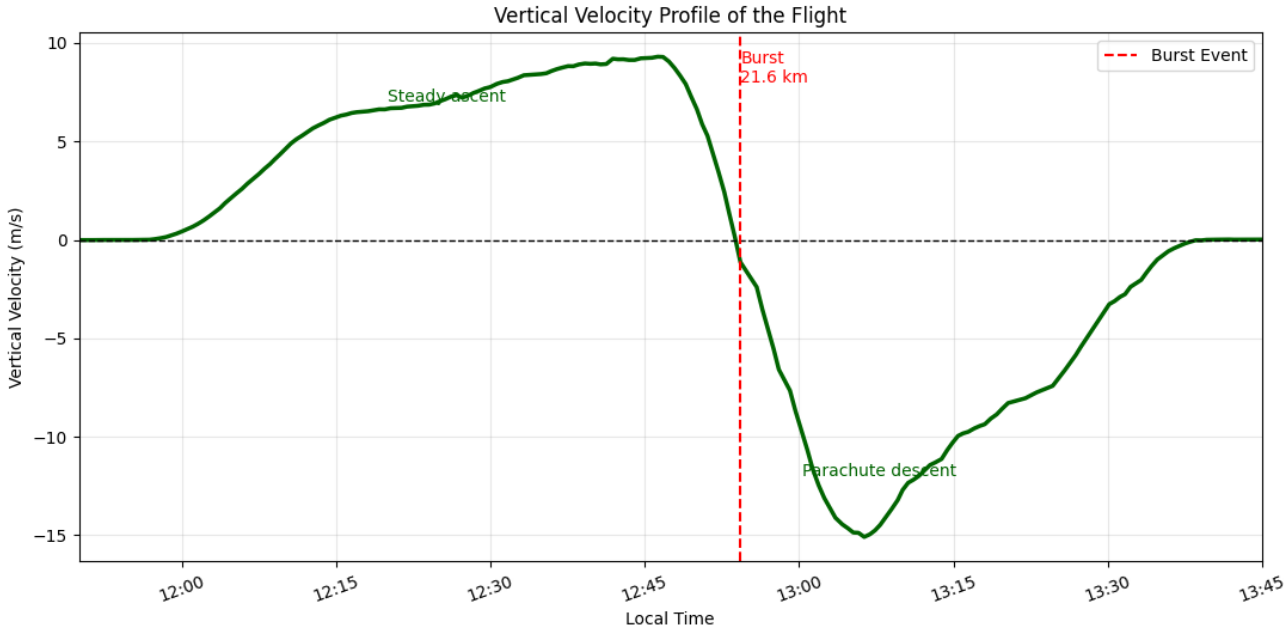


Figure 35. Vertical velocity profile.

The vertical velocity profile was obtained by differentiating the altitude telemetry with respect to time and provided more insight into the behaviour of the payload throughout the mission. The resulting profile plot clearly shows the major flight phases, including ascent, burst, rapid descent, parachute stabilisation, and landing. During ascent, the payload seemed to have a relatively stable positive vertical velocity throughout the majority of the climb phase. Analysis of the telemetry indicated an average ascent velocity of approximately 4–5m/s, with peak ascent velocities approaching 9–10m/s immediately prior to burst. These values were pretty consistent with expected flight characteristics for this flight configuration.

The burst event is clear within the velocity profile as a rapid transition from positive ascent velocity to negative descent velocity. Immediately after the balloon burst, the payload experienced a short period of accelerated descent before the parachute stabilised the system and reduced the speed of decent. Following parachute deployment, the payload descended at a lower and more stable terminal velocity of approximately 5m/s, suitable for safe recovery operations. The relatively smooth descent behaviour following stabilisation suggested successful parachute deployment and stable aerodynamic performance during the recovery phase. Overall, the measured vertical velocity behaviour was consistent with predicted high-altitude balloon flight dynamics and provided clear evidence of nominal ascent and recovery performance throughout the mission.

3.5.2 Vertical Acceleration Analysis

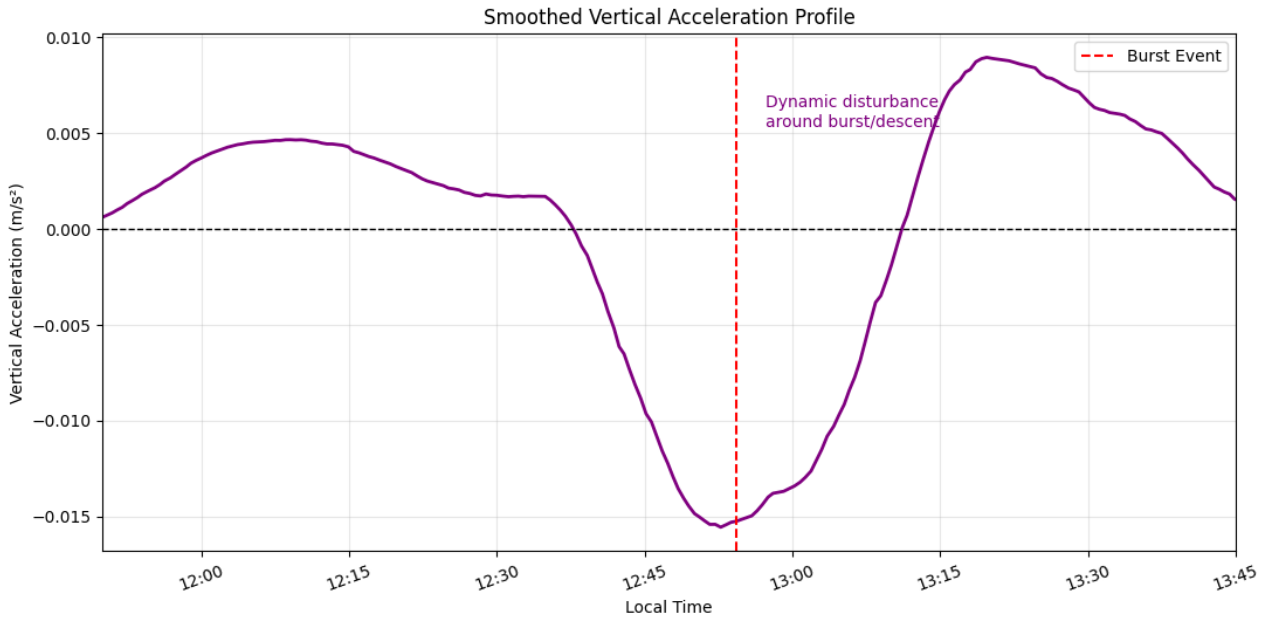


Figure 36. Vertical acceleration profile.

The vertical acceleration plot was estimated from the derivative of the smoothed velocity data and was used to identify specific events during flight. Due to the relatively low temporal resolution and noise associated with GPS altitude measurements, the acceleration data was interpreted qualitatively rather than quantitatively. The largest acceleration disturbance occurred around the balloon burst moment, where the payload rapidly transitioned from ascent to descent. Smaller fluctuations during ascent and descent were likely associated with atmospheric turbulence, payload oscillation, and parachute deployment dynamics. Following burst, the acceleration profile indicated that the payload gradually stabilised into a controlled parachute descent before landing. These events are important because they may influence sensor stability and internal payload behaviour during flight.

3.6 Thermal Sensor Results

The MLX90640 infrared thermal sensor successfully recorded thermal evolution data during the early stages of ascent and provided a qualitative view of the internal thermal behaviour of the payload throughout flight. The recovered thermal data showed gradual cooling within the observation chamber as the payload ascended into the colder upper atmosphere, with noticeable spatial temperature variations across the sensor field of view. Warmer regions remained visible around the insulated interior and avionics section during the earlier stages of flight, suggesting that the internal insulation, fire blanket material, and handheld heating pads definitely helped retain heat within the payload structure.

As ascent continued, the thermal imagery showed an overall reduction in observed temperatures and increasing thermal isolation within the inside of the chamber, which is

consistent with prolonged exposure to low external temperatures. However, because the payload experienced premature power loss before reaching the planned trigger altitude, no clear flash evaporation event, rapid cooling spike, or plume-like thermal structure was observed within the recorded data. Despite this limitation, the thermal sensor data still provided useful insight into the internal thermal evolution of the payload during ascent and demonstrated that the MLX90640 system remained operational prior to the shutdown event.

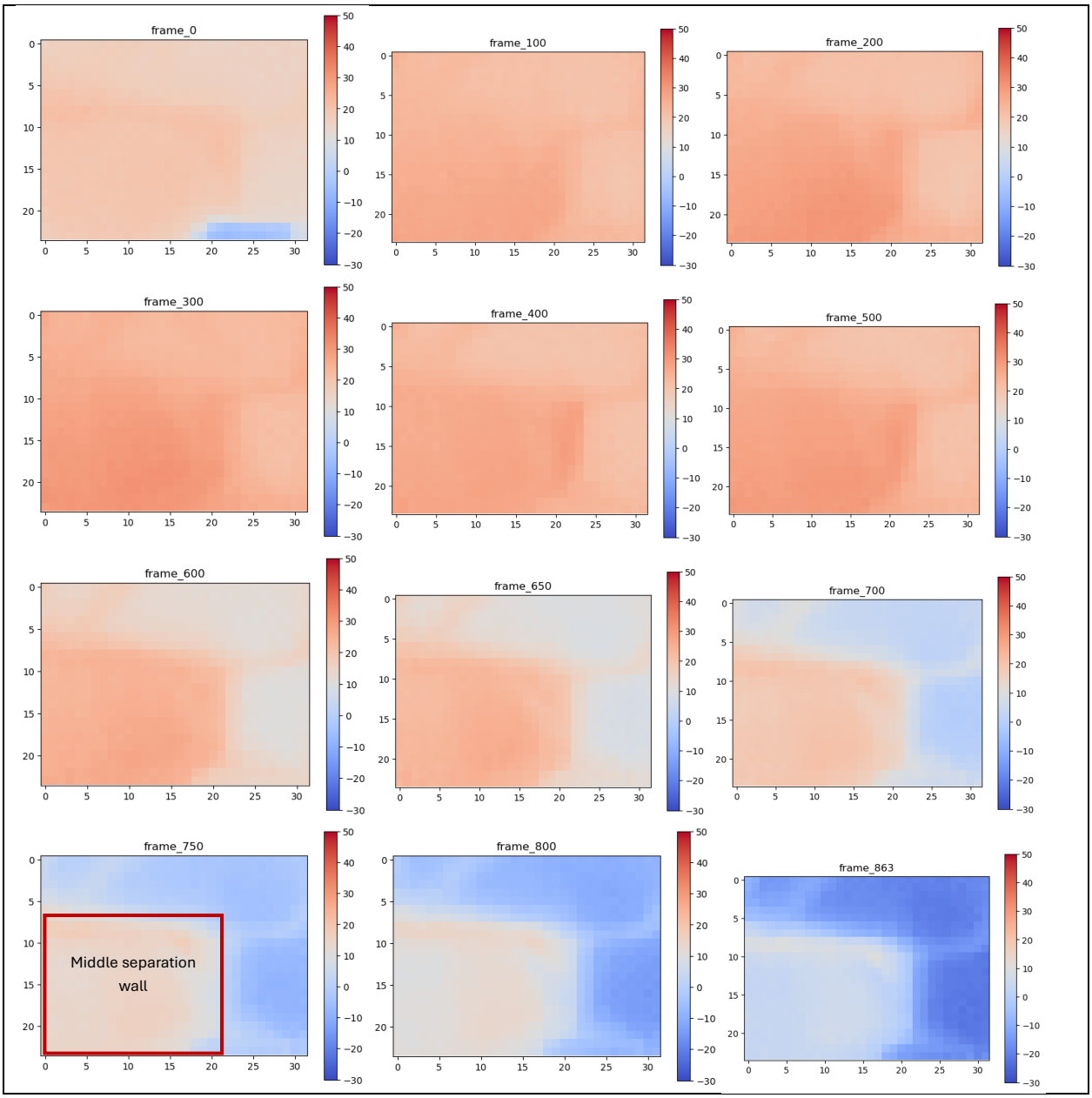


Figure 37. Time Series displaying Thermal Evolution of Low Pressure Chamber. Note frames 600-863 show contrast colours of middle separation wall and outter tupperware palstic regions which have more exposure to lower temperatures.

3.7 Discussion of results; Comparison with previous missions or literature.

A major aim of the SQRT mission was to investigate whether flash evaporation of water could be observed during high-altitude balloon flight under low-pressure stratospheric conditions. The experiment was designed around the idea that once the payload reached pressures of 35mbar, liquid water released into the chamber would rapidly flash-evaporate, creating a visible vapour plume followed by freezing. The flight data was therefore analysed to determine whether the observed pressure and thermal behaviour matched these expected physical processes. The results only partially matched the expected physics behind the experiment. SQRT lost power during ascent before reaching the planned trigger altitude, meaning the experiment did not fully occur in flight. From the available telemetry, no clear rapid temperature drop or plume event was observed before shutdown. Although the wider flight telemetry showed that the balloon itself eventually reached the required low-pressure environment, the SQRT payload was no longer operational by that stage. As a result, the experiment could not directly confirm the expected flash evaporation behaviour or compare the thermal response against the original model predictions. Despite this, the pressure and altitude profiles were still broadly consistent with the expected stratospheric environment used during mission planning. This suggests that the overall atmospheric model and trigger threshold were reasonable, with the main issue being payload survivability rather than the underlying physics itself. This links back to the original scientific motivation, but in bit of a limited way... The mission was inspired by processes such as plume formation on icy moons like Enceladus and Europa, where liquid or volatile material may be exposed to very low-pressure environments and rapidly change phase. SQRT did not reproduce the full complexity of these planetary systems, but it aimed to test one smaller part of the same physics: how water thermally responds when suddenly exposed to reduced pressure. Because the trigger event didnt happen, the flight data cannot be directly compared to plume formation theory or flash evaporation cooling predictions. However, the wider flight telemetry did show that the balloon reached the type of low-pressure environment needed for this behaviour to be studied. This means the mission idea itself was still scientifically reasonable, even though the payload did not remain powered-on long enough to collect the intended thermal data. The same limitation applies to the engineering motivation, such as flash desalination and reduced-pressure evaporation systems, where the relevant physics depends on observing the actual phase-change event.

Compared with many previous balloon or TupperSat-style missions, SQRT placed a much stronger emphasis on using the stratospheric environment itself as part of the experiment rather than simply as a transport platform. Many student high-altitude balloon missions focus primarily on atmospheric profiling, imaging, telemetry demonstrations, or environmental sensing. In contrast, SQRT was designed around a transient physical process that could only occur under specific low-pressure stratospheric conditions. This made the mission more reliant on the surrounding flight environment and introduced additional operational complexity. The experiment relied on reaching a narrow pressure regime before triggering the release system, making the mission significantly more sensitive to timing, power stability, and environmental survivability than standard telemetry or imaging payloads.

The planned flash evaporation event also represented a much shorter-duration and higher-risk experimental process compared with more passive atmospheric measurement missions.

Although the experiment did not fully complete due to premature payload shutdown, the mission still demonstrated a more ambitious systems-level integration between environmental conditions, trigger logic, thermal behaviour, and payload operation than many previous student balloon experiments highlighted on Brightspace. In this sense, SQRT represented a more experimentally driven use of the high-altitude balloon platform rather than simply a method of carrying instruments into the stratosphere. One of the main differences between the ideal experimental model and the real flight data was the influence of the stratospheric flight environment itself. In laboratory conditions, the experiment would have occurred in a relatively controlled and stable system, whereas the balloon flight introduced continuous motion, changing thermal conditions, vibration, and varying ascent dynamics. These effects probably influenced both the payload behaviour and the quality of the recorded telemetry. During flight, the payload was exposed to oscillation, rotation, and wind-driven disturbances which wouldn't normally happen during controlled ground testing. These effects may have contributed to fluid redistribution within the chamber, sensor noise, and variations in thermal response. In addition, changing ascent rates and external temperatures introduced continuously evolving environmental conditions rather than the fixed-pressure conditions typically assumed during simplified modelling. Thermal shielding and insulation effects may also have altered the behaviour of the payload compared with expectations. While insulation was intended to protect the water and electronics from extreme cold, it may also have delayed thermal response times or contributed to uneven internal heating, not to mention of course; potentially frying the power-pack!

Although these real-world effects complicated interpretation of the data, they also highlighted the value of performing the experiment in an actual stratospheric environment rather than under purely laboratory conditions. The mission demonstrated that high-altitude balloon experiments involve complex environmental and engineering effects that are difficult to fully reproduce during ground testing alone. It also provides a unique and low cost low pressure and thermal environment, perfectly suited to the original objectives of the SQRT mission.

4. PERSONAL REFLECTION

Overall, I think one of the strongest parts of the SQRT project was how much the mission evolved and matured over the course of the full semester. Looking back at the original idea compared to the final flight-ready payload, the difference is honestly kind of ridiculous, but in a positive way. Early on, the project was much more focused on the "cool science idea" side of things, whereas later stages forced us to think more realistically about systems engineering, integration, operations, testing, and actual payload survivability. I think one of the things the team did well was gradually shifting from "wouldn't this be an interesting idea?" to "can this actually survive being launched into the stratosphere and still work without freezing or breaking?" which turned out to be a much harder question. Personally, I think one of my stronger earlier contributions of the mission was helping bridge the science and engineering



sides of the project having led the research on the desalination and icy-moon plume research. Thereafter, I believe I was helpful on the hardware elements of the actual experiment having arranging procurement of all materials and collaborating with Lab Technicians at the School of Medicine to donate the necessary syringes and tubing. I also designed and printed the twistable syringe cap, the electronic bay cover, the compartment separation wall and the payload spacing separators for the threaded walls. During the later testing and verifaction stages of the mission I took lead on documentation writing and worked mostly on the OUM, the presentation slides and contributed on the Verification Control Document and Tracking Matrix. One key responsibility that I took very seriously was our Risk Assessment especially during the later stages where the payload design had to keep adapting around practical limitations. My main worry of the mission was premature water freezing within the chamber and only when extra payload insulation was introduced during testing did the payload prove mitigated. Unfortunately, I may have taken this too far as it's clear that the internal temperature of the payload may have been much too hot and could be the cause of the power loss. I did aim to involve myself where possible in software configuration too, but I believe that the team had that under smooth control and as such there were limited tasks available. That said, the software team knew they always had help available should they need it.

I think the team worked well under pressure overall, even if the pressure levels became mildly concerning at certain points. The amount of documentation, review preparation, and design iteration between the MDD, CDR, and FAR stages was much larger than I originally expected, but it definitely improved the project in the long run. The review process forced us to justify design choices properly rather than just relying on 'well I hope it works' assumptions, and by the final stages the payload felt much more thought-out than it had initially. Things like the risk register, operational procedures, telemetry planning, and launch-day preparation all became far more detailed over time, and I think that made a big difference during the actual flight phase of the mission.

Launch day itself was also one of the most bitter-sweet yet satisfying parts of the whole project. Despite all the stress beforehand, the balloon launched successfully, telemetry was received, the flight profile looked physically reasonable, and the payload was ultimately recovered fully intact. Even though SQRT shut down before completing the planned trigger event, it still felt rewarding seeing an actual system we had designed and built operating in a real flight near-space environment rather than just sitting on a lab bench in Beech Hill as it had been for months prior. It was particularly bitter-sweet for me as I mainly contributed to the actual experimental bus rather that the sensor and electronic configurations, so to know that it was so close to completing the actual experiment and obtaining thermal evolution data and maybe even a flash-evaporation plume only for my over-heating antics to potentially be the cause of that not happening was a bit of a let-down! That said, as mentioned, I took my role in ensuring the water stayed in liquid form seriously and due to my suggestions and insulation recommendations the payload was absolutely kept warm enough to avoid freezing, so that's a small but unlucky success.

That being said, there were definitely quite a few things that did not go so well, and most of them only became obvious once deadlines started collapsing together near the end of the semester. The biggest issue was probably underestimating how much time integrated



testing, troubleshooting, and documentation would actually require. Early in the project it felt like there was loads of time, but near FAR and launch preparation everything suddenly became interconnected. Small design changes started creating knock-on effects elsewhere in the payload, and because some decisions were still evolving relatively late into the process, it became harder to properly verify everything before flight. In hindsight, the project probably stayed in "active redesign mode" for too long before reaching a proper finalized design. There was also a tendency at times to focus heavily on getting subsystems individually working rather than repeatedly testing the entire payload as one complete integrated system under realistic conditions. The SQRT shutdown itself was obviously the clearest example of this. During bench testing, the payload appeared functional, but the real stratospheric environment introduced factors that were much harder to replicate properly on the ground. Vibration, changing temperatures, pressure variation, power behaviour, and long-duration operation all became much more important than expected once the payload was actually flying. Looking back now, I think we probably underestimated how unforgiving real flight conditions can be compared to controlled lab testing. There were also moments throughout the semester where communication and workload balancing could have been better, particularly when multiple tasks became dependent on each other near deadlines. A lot of the project ended up becoming reactive rather than proactive simply because there was so much happening simultaneously. At several points the engineering strategy was probably best described as "this should hopefully be fine," which is not always the most reassuring sentence in aerospace projects especially with all of the effort being put in by the team.

I think the biggest thing I learned from this project overall was how important systems engineering discipline actually is once a project becomes even moderately complicated. At the start of the semester, things like requirements tracking, risk analysis, configuration management, and operations procedures honestly felt a bit excessive for a student balloon project. I even remember mentioning this as a joke to Caimin (instructor) that I thought maybe we were doing too much for a 'fake space test' to which he responded 'This is to get you ready for ESA and the real-world, you should be taking it seriously'. He was absolutely right - By the end, I understood much more clearly why those processes exist in real engineering programmes. One small overlooked issue can very quickly propagate through an entire system, especially when power, thermal behaviour, telemetry, triggering logic, and environmental survivability are all linked together. I also learned that testing late is almost always a mistake. The phrase "we'll verify that later" usually translates into "this will become a problem at the worst possible moment… who knows maybe even in flight?" If I was repeating the project, which I would love to do, I would push much harder for earlier payload procurement, design and building, in order to have much earlier testing and more realistic environmental testing suited to the actual launched payload, stricter design freeze deadlines, and clearer assumptions about what actually counts as "flight ready."

I also think the project changed the way I approach technical work in general. Earlier in the semester I was probably more focused on whether an idea was theoretically interesting and do-able whereas now I think much more about whether it is operationally robust and realistically testable. One of the biggest personal improvements for me was becoming more comfortable interpreting messy real-world telemetry data through bith this module and

others rather than expecting everything to behave perfectly like it seems to in movies. Real flight data is clearly noisy, incomplete, and sometimes plain annoying, but learning how to extract useful conclusions from these datasets was probably one of the most valuable parts of the entire experience. Overall, even though the mission did not fully achieve the original experimental objective, it still felt like a genuinely valuable engineering project because it exposed all the complicated, slightly chaotic realities that real aerospace projects have to deal with. Honestly, the fact that things went wrong at times probably taught me more than if everything had worked perfectly first try. Again…bitter-sweet, but a real lesson!

4.1 Lessons Learnt

4.1.1 What would you have done differently? Why?

If I were repeating the project, I would place much more emphasis on earlier integrated testing of the complete payload rather than testing individual subsystems separately for most of the semester. A lot of components appeared to work correctly on the bench, but the real problems only became obvious once everything was running together under more realistic conditions such as the 'Fridge Tests'. In particular, I think more environmental and endurance testing would have helped identify issues with power stability and long-duration operation before flight. I would also push for a firmer design earlier in the project timeline. There were several late-stage changes that improved parts of the payload, but they also reduced the amount of time available for full testing, verification and troubleshooting. Looking back, having a slightly simpler but fully tested system would probably have been more valuable than continuing to optimise individual features too close to the launch windows.

Personally, I also think I underestimated how important systems engineering discipline becomes once a project starts getting more complicated. Early in the semester I was much more focused on the scientific side of the mission, whereas by the end I realised how important requirements tracking, risk management, documentation, and operational planning actually are. The project definitely changed how I think about engineering work, especially in terms of testing assumptions rather than relying on them.

4.1.2 What could we have done differently?

As a team, we probably could have started the more hand-on work such as design, printing and then testing and verification much earlier in the semester. A lot of work became concentrated towards the time of the Tenerife trip, the FAR stage and launch preparation, which created additional pressure once multiple subsystems (and other semester modules) became dependent on each other. Earlier system-level testing would probably have improved confidence and reliability of the payload before launch. The project may also have benefited from simplifying some parts of the payload design earlier on. The experiment concept was ambitious and scientifically interesting, but it introduced several dependencies between triggering logic, thermal behaviour, telemetry continuity, and environmental survivability. Simplifying some of these interactions may have reduced mission risk and improved overall reliability.

Communication and workload balancing could also have been improved at certain stages of the project, particularly when deadlines became compressed. At times the team became reactive rather than proactive simply because there were so many parallel tasks occurring simultaneously. More structured planning and clearer task ownership earlier in the semester would likely have helped reduce this. I feel partially response for this as project manager but I do feel I expressed openly that no member of the team should feel isolated in their work and would always have help available if needed. I do think I could have contributed more across the electronics and software tasks but I also think that some members of the team may have pushed back against this with the mindset of 'too many cooks in the kitchen'. Despite these challenges, the project still achieved most of its operational goals and provided valuable practical experience in systems engineering, payload integration, telemetry analysis, and flight operations. Even though the experiment didn't happen as intended, the issues encountered during the mission ultimately became some of the most useful learning experiences from the project overall.

5. CONCLUSIONS

The SQRT mission aimed to investigate the behaviour of water under low-pressure stratospheric conditions using a high-altitude balloon platform. The payload was designed to trigger a flash evaporation event at approximately 35 mbar in order to study rapid phase change, latent heat cooling, and the resulting thermal evolution of the system.

Although the payload experienced premature power loss before reaching the intended trigger conditions, the mission still provided valuable engineering and scientific insight into operating experimental systems within a real stratospheric environment. Analysis of the flight telemetry showed that the balloon successfully reached the expected high-altitude flight regime, with pressure, temperature, ascent, and descent profiles broadly consistent with predicted high-altitude balloon behaviour. The reconstructed flight profile clearly demonstrated the major phases of the mission, including steady ascent, balloon burst, rapid descent, parachute stabilisation, and recovery. The measured flight dynamics and environmental conditions also supported the validity of the original atmospheric modelling used during mission planning. While the primary scientific objective could not be fully achieved due to the early shutdown of SQRT, the mission highlighted several important engineering lessons regarding payload survivability, systems integration, environmental testing, and operational robustness. In particular, the project demonstrated how strongly real flight conditions can influence payload behaviour compared with controlled laboratory assumptions. Overall, the project provided very valuable practical experience in systems engineering, telemetry analysis, mission operations, and experimental design. Despite the incomplete experimental outcome, SQRT successfully demonstrated the challenges and complexities of performing science-driven experiments within the stratospheric environment and provided a strong foundation for future development and refinement of similar payload systems.



REFERENCES & APPENDICES

Research Papers and Literature

- Behnken, B. and Stoffregen, T. (2017) 'High-altitude ballooning for scientific and educational payloads', *Journal of Aerospace Engineering*, 30(4), pp. 1–12.
- Ingersoll, A.P. and Ewald, S.P. (2011) 'Total particulate mass in Enceladus plumes and mass of Saturn's E-ring inferred from Cassini ISS images', *Icarus*, 216(2), pp. 492–506.
- Raja, M.M.A., Murugavel, K.K., Rajaseenivasan, T. and Srithar, K. (2016) 'A review on flash evaporation desalination', *Desalination*, 57, pp. 13462–13471.
- Matson, D.L., Castillo-Rogez, J.C., Davies, A.G. and Johnson, T.V. (2012) 'Enceladus' plume: Compositional evidence for a hot interior', *Icarus*, 221(1), pp. 53–62.
- Spencer, J.R. et al. (2006) 'Cassini encounters Enceladus: Background and the discovery of a south polar hot spot', *Science*, 311(5766), pp. 1401–1405.

Web References for Figures/Maps

- **SondeHub trajectory prediction for flight path.**
  SondeHub (2026) *SondeHub Predictor*.
  Available at: SondeHub Predictor
  (Accessed: 23rd April 2026).
- **Armagh Observatory weather data for site forecast data**
  Armagh Observatory (2026) *Climate and Weather Data Archive*.
  Available at: Armagh Observatory Climate Archive
  (Accessed: 19th May 2026).

Appendix Note;

- All code used for this report's telemetry processing, calculations, and figure creation is available in the this GitHub repo along with the associated mission datasets.

Statement of Authorship;

This report was written independently and reflects my own analysis and interpretation of the SQRT mission data. The mission itself was a collaborative project completed by the SQRT team, consisting of Orlaith Ní Dhúill, Javier Ayuso, Caimin Keavney, Bruce Ritter, and Elyana Abdallah. Additional telemetry and supporting flight data used within this report were kindly provided by the SkyNet team, consisting of Fazira Azzahra, Casey Donovan, Abin Dominic, Daragh Haughton, and Liam Moran. Special thanks are also given to module coordinator Lorraine Hanlon, lecturer Andrew Kirwan, and teaching assistants Caimin Mac Cionnaith and Cuán De Barra for their support throughout the mission and report process.

AI-assisted tools, including Claude and Gemini, were used to support aspects of the data processing, coding, report structuring, and analytical workflow presented within this report.